

Table 2: 2015 Energy Usage Intensity Benchmarks

Energy Type	Energy Usage Intensity (kBtu/ft ² -yr)	Energy Usage Intensity (kBtu/rm-yr)	Energy Cost Intensity (\$/ft ² -yr)	Energy Cost Intensity (\$/rm-yr)
Electricity	51.0	89,509	\$2.13	\$3,729
Natural Gas	55.5	97,353	\$0.37	\$652
Total	106.5	186,862	\$2.50	\$4,381

The subsequent table shows cost and water usage benchmarking metrics. The abbreviation of “kgal” represents 1,000 gallons of water. In 2015 the hotel used 27,706,350 gallons of water.

Table 3: 2015 Water Usage Summary

Annual Water Consumption (kgal/yr)	Annual Water Cost (\$/yr)	Water Usage Intensity (gal/ft ² -yr)	Water Usage Intensity (gal/rm-yr)	Water Cost Intensity (\$/ft ² -yr)	Water Cost Intensity (\$/rm-yr)
27,706	\$326,760	55	97,215	\$0.65	\$1,147

Summary of Available Energy Conservation Measures

Twelve Energy Conservation Measures (ECMs) are presented in this report. If all of the ECMs were completed, the hotel would save about \$138,400 per year with an investment of \$939,000. With incentives of \$245,850 the aggregate simple payback becomes 5.0 years. The following table shows a summary of the annual cost savings and paybacks expected for each of the measures.

Table 4: Energy Conservation Measures

ECM	Description	Total Cost Savings (\$/yr)	Project Cost (\$)	Incentive (\$)	Simple Payback (yrs)
1	Chiller Plant Replacement	\$43,000	\$610,000	\$90,900	12.1
2	Occupancy Based Guest Room Thermostat Control	\$36,000	\$120,000	\$75,900	1.2
3	VAV AHU Discharge Air Temperature and Static Pressure Resets	\$15,000	\$20,000	\$9,600	0.7
4	Low-Flow Showerheads	\$9,800	\$25,000	\$0	2.6
5	Variable Kitchen Exhaust and Make-Up	\$7,800	\$40,000	\$18,900	2.7
6	Variable Volume Chilled Water Pumping	\$6,900	\$42,000	\$16,200	3.7
7	Single Zone AHU Variable Volume Conversion	\$6,900	\$36,000	\$14,700	3.1
8	EC Motor Installation in Walk-In Coolers	\$6,200	\$31,000	\$14,400	2.7
9	Condenser Water Temperature Reset	\$5,300	\$2,000	\$1,500	0.1
10	Variable Volume AHU-3 Conversion	\$800	\$8,000	\$2,100	7.4
11	Variable Speed Pool Pumps	\$600	\$4,000	\$1,500	4.2
12	Occupancy Sensors for Lighting Control	\$100	\$1,000	\$150	8.5
Total		\$138,400	\$939,000	\$245,850	5.0